30-Day Readmission Prediction for Diabetic Patients Using Machine Learning

Bui Vi Anh, Nguyen Ngo Anh Thu, Nguyen Thai Anh Thu

National Economics University – Class DS66B (Group 13)
Supervisor: Dr. Luong Van Thien

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Outline

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2. Existing Methodologies
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4. Experiments
5. Conclusion

Problem Statement

Challenge: Predicting whether a diabetic patient is readmitted within 30 days of discharge

Key Issues:

- Severe class imbalance: only about 11% positive encounters
- Multiple encounters per patient: data-leakage risk across splits
- Heterogeneous, high-cardinality data with heavy missing values
- Weak predictive signal: no single feature is strong (max Pearson $r = 0.167$)
- Threshold choice and operational constraints
- Fairness across patient subgroups

Solution: Leakage-safe pipeline combining feature engineering + CatBoost + validation-selected threshold

Scope: risk stratification to prioritise follow-up care, not individual diagnosis or a clinically validated tool

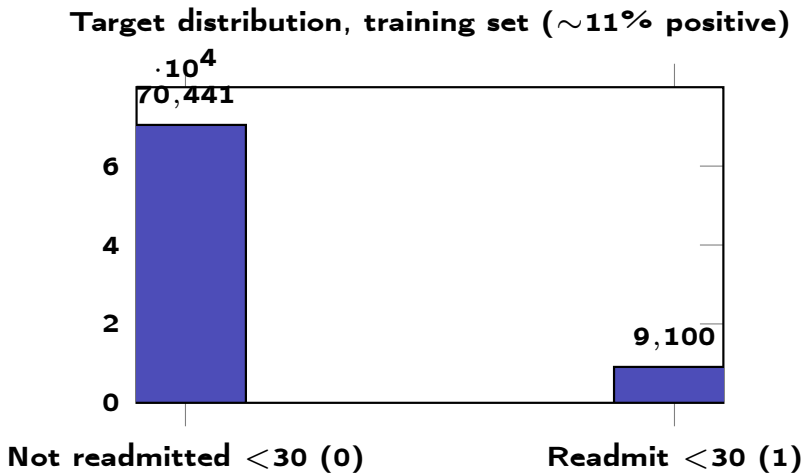
Dataset: Diabetes 130-US Hospitals

- UCI repository, 1999–2008, 130 US hospitals
- 101,766 encounters, 50 columns
- 71,518 patients; 16,773 (23.5%) have more than one encounter
- Removed death/hospice: 2,423 $\Rightarrow$ **99,343** encounters

Target: $y = 1$ if readmitted < 30 days

Table 1: Original target distribution

readmitted	Count	%
NO	54,864	53.9
> 30	35,545	34.9
< 30	11,357	11.2



Challenges in the Data

Imbalance and leakage

- "Predict no for all" gives 88.8% accuracy, 0 detections
- ≈ 8 negatives per positive
- Same patient in train and test inflates results $\Rightarrow$ split by `patient_nbr`

Messy features

- `diag_1/2/3`: 717/749/790 ICD-9 codes
- Missing ("?"): weight 96.9%, specialty 49.1%, payer 39.6%, race 2.2%
- Lab "None" = not tested (94.7% glucose, 83.3% A1C), not missing
- Many near-constant medication columns

Weak Signal: Exploratory Analysis

Pearson r with target

`number_inpatient`	0.167
`number_emergency`	0.073
`number_diagnoses`	0.054
`time_in_hospital`	0.046

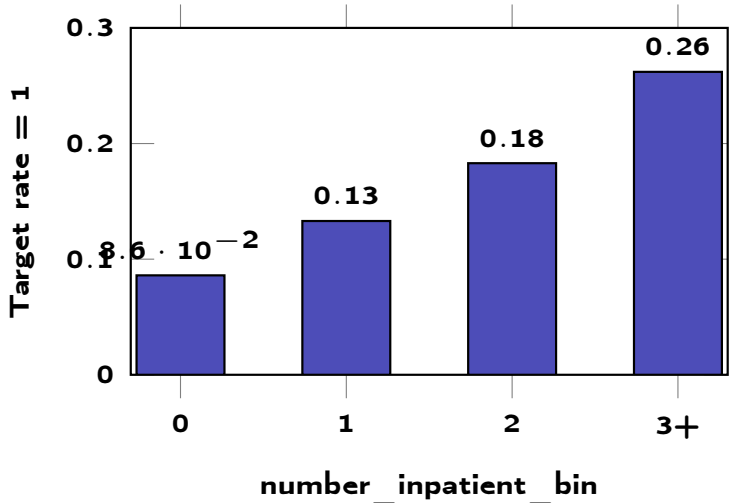
Cramér's V

`discharge_disposition_id`	0.112
`diag_3_group`	0.044
`insulin`	0.043

All other categorical $V < 0.045$; gender not significant ($p = 0.516$).

Readmission rate by prior inpatient visits

Prior inp.	0	1	2	3+
Rate (train)	8.6%	13.3%	18.3%	26.2%



Insights: No single feature is strongly related to early readmission; prior inpatient visits and discharge disposition are the most informative. Signal is weak and spread across many variables.

Related Work Overview

Dataset and typical approaches:

- Strack et al. (2014): dataset assembled from Health Facts, now a standard benchmark
- Emi-Johnson & Nkrumah (2025): LR, RF, XGBoost, DNN; XGBoost best, SHAP for interpretation
- Shang et al. (2021): ML classifiers on a Cerner-derived set; prior inpatient visits influential
- Reviews (Kansagara 2011): administrative data gives modest discrimination

Why reported results vary so much:

- AUROC from low 0.60s up to 0.95–0.97 (EmbPred30: 97.4%; Goudjerkan & Jayabalan: ≈95%)
- Likely causes: encounter-level random splits (23.5% of patients have repeat encounters), outcome-encoding features (death/hospice), threshold tuned on test
- Split protocol often not stated ⇒ numbers not directly comparable

Our Contribution: methodological rigor, namely patient-grouped split, AP-based selection, validation threshold, bootstrap CI and calibration analysis

Traditional and Ensemble Methods

Single Learners:

- **Logistic regression**: linear on log-odds, clinical reference
- **Decision tree**: non-linear, interpretable, noisy if deep

Ensembles:

- **Bagging** (random forest): reduces variance
- **Boosting** (CatBoost): sequential, reduces bias and variance

Trade-off: bagging is safe and low-variance; boosting can find more structure but needs tuning on weak, noisy signal. We measure it, not assume it.

Gradient Boosting and CatBoost

Additive model fitted on the negative gradient of the loss:

$$F_m(x) = F_{m-1}(x) + \eta \cdot h_m(x), \qquad p = \frac{1}{1 + e^{-F(x)}} \tag{1}$$

For logistic loss the pseudo-residual is $y_i - p_i$.

CatBoost:

- Symmetric (oblivious) trees: fast, mild regularisation
- Ordered boosting to avoid prediction shift
- Categorical columns one-hot encoded $\Rightarrow$ native handling **not** used (fair comparison)

Loss Function and Class Imbalance

Logloss:

$$L = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log p_i + (1 - y_i)\log(1 - p_i)\right] \tag{2}$$

Class weights

- $w_c = N/(K \cdot N_c)$: ≈ 4.4 (pos), 0.56 (neg)
- Probabilities no longer match prevalence

Threshold moving

- Loss unchanged
- Probabilities stay honest

Statistical tools: chi-square, Cramér's V, patient-cluster bootstrap CI, Holm correction, ECE, decision curve, cost threshold $t^* = \frac{1}{1+r}$.

Proposed Method

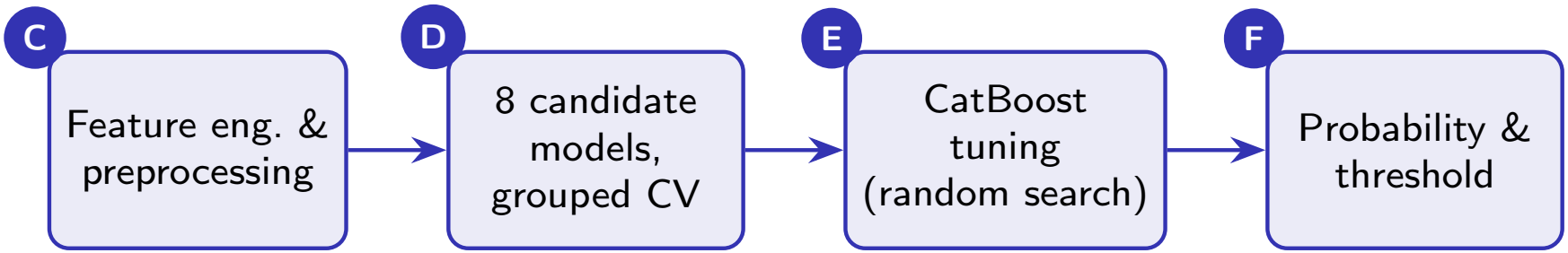


Figure: Overall pipeline with four phases: (C) Feature Engineering and Preprocessing, (D) Candidate Models, (E) Final Classifier, (F) Probability and Decision Threshold

Pipeline Architecture

Key Components:

1. **Feature Engineering and Preprocessing**: ICD-9 grouping, 7 engineered features, robust scaling, one-hot
2. **Candidate Models**:
   - LR, DT, RF, CatBoost, each with/without class weight
   - Compared by patient-grouped CV with AP
3. **Final Classifier**:
   - CatBoost tuned by randomised search
   - Refitted on full training set
4. **Decision Threshold**: max-F1 on a patient-disjoint validation split, fixed before the test set

Data Cleaning and Missing Values

Cleaning Process:

- Exclude death/hospice discharges; merge "unknown" admin codes
- Drop `weight`; `race`, `payer_code`, `medical_specialty` $\rightarrow$ "Unknown" (missingness informative for 2 of 3 by chi-square)
- `max_glu_serum`, `A1Cresult`: "None" $\rightarrow$ "Not tested"
- **ICD-9 grouping**: 700+ codes $\rightarrow$ broad groups (`diag_*_group`)
- Drop 15 of 23 near-constant medication columns (top value $\geq$ 99%)
- Winsorize 5 count variables at 1st/99th percentile (train only)

Principle: every data-dependent statistic is fitted on the training set only.

Feature Engineering

Feature	Definition
`total_previous_visits`	outpatient + emergency + inpatient
`has_prior_inpatient / emergency`	1 if count > 0
`medications_per_day`	`num_medications` / max(`time_in_hospital`, 1)
`labs_per_day`	`num_lab_procedures` / max(`time_in_hospital`, 1)
`procedures_per_day`	`num_procedures` / max(`time_in_hospital`, 1)
`treatment_intensity`	labs + procedures + medications

- Row-wise only, no target, no cross-row statistics $\Rightarrow$ no leakage
- 31 original + 7 new = **38 predictors** (15 numeric, 23 categorical)
- RobustScaler for numeric, One-Hot (`handle_unknown="ignore"`) for categorical

Patient-Level Data Splitting

Split	Encounters	Patients	Positives	Rate
Train (all)	79,541	–	–	11.44%
fit	63,807	–	–	11.43%
validation	15,734	–	1,802	11.45%
Test	19,802	13,998	2,216	11.19%

- `GroupShuffleSplit` by `patient_nbr`, 80/20, seed 42: **0 shared patients**
- CV: `StratifiedGroupKFold` (3 folds for comparison, 5 for tuning)
- Validation = 20% of training patients; the split is grouped but not stratified (rates already match)
- Test set used **once**, after model, hyperparameters and threshold are fixed

Final Classifier and Decision Threshold

CatBoost tuning

- `RandomizedSearchCV`: 20 configs $\times$ 5 grouped folds, AP score
- Search: depth 4–9, lr 0.01–0.21, L2 1–11, iterations 200–799
- Best: depth 5, 764 iter., lr 0.058, L2 3.31
- CV AP $= 0.234$

Threshold

- Fit on the fit set, predict validation
- Maximise F1 on precision-recall curve $\Rightarrow$ **0.145**
- Fixed, then applied to the test set

No class weights in the final model: imbalance is handled by the threshold. The threshold comes from a model fitted on the fit set (63,807) and is applied to the model refitted on all training rows (minor limitation).

Experimental Setup

Dataset: Diabetes 130-US Hospitals

- 99,343 encounters after cleaning, split by patient
- Python 3.12, scikit-learn, CatBoost, seed 42, CPU

Table 2: Naive reference policies (test set)

Policy	Acc.	Recall	F1	AP	AUC
Flag nobody	0.888	0.000	0.000	0.112	0.500
Flag everybody	0.112	1.000	0.201	0.112	0.500
Tuned CatBoost (0.145)	0.762	0.431	0.288	0.244	0.677

Baselines: LR, DT, RF, CatBoost (each with/without class weight) = 8 candidates; untuned CatBoost is the reference for tuning

Metrics: AP (primary; random = prevalence 0.112), ROC-AUC, Recall, Precision, Specificity, F1/F2, MCC, Brier, ECE. Accuracy is misleading here.

Model Selection Results

Table 3: 3-fold grouped CV (fit set) and validation results

Model	CV AP	Val AUC	Val AP
CatBoost	**0.227 ± 0.005**	**0.678**	**0.233**
Random Forest	0.219 ± 0.004	0.675	0.221
CatBoost + class weight	0.216 ± 0.002	0.669	0.225
Random Forest + class weight	0.211 ± 0.001	0.670	0.207
Logistic Regression	0.209 ± 0.003	0.665	0.213
Logistic Regression + class weight	0.207 ± 0.003	0.667	0.212
Decision Tree	0.192 ± 0.001	0.648	0.212
Decision Tree + class weight	0.190 ± 0.002	0.644	0.205

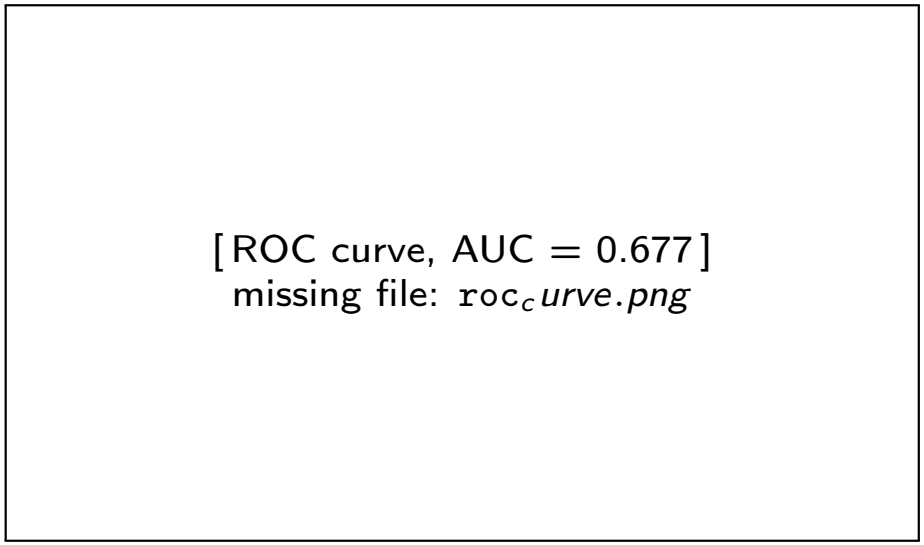
Key Results:

- CatBoost first on both CV and validation (about 2× random AP)
- Gap to random forest (0.008) is comparable to fold std (0.005): CV alone does not prove superiority; CV and validation together support the choice
- All validation AUCs in 0.644–0.678: weak-signal setting

Final Performance on the Test Set

Table 4: Patient-cluster bootstrap 95% CI

Metric	Est.	95% CI
ROC-AUC	0.677	0.663–0.691
AP	0.244	0.221–0.268
Recall	0.431	0.404–0.458
Precision	0.217	0.201–0.232
Specificity	0.804	0.796–0.811
F1	0.288	0.270–0.306



Key Results:

- AP = 2.18× random; flags 22.2% of *encounters*, detects 43.1% of readmissions
- Confusion matrix: TP 954, FN 1,262, FP 3,447, TN 14,139; lift 1.94, MCC 0.178
- NPV 0.918 vs 0.888 by chance: "not flagged" gives little reassurance

Comparison with Baselines

Table 5: Paired patient-cluster bootstrap (2,000 replicates), Holm-corrected; differences = tuned CatBoost − competitor

Model	AUC	AP	ΔAUC [95% CI]	ΔAP [95% CI]	Signif.
Tuned CatBoost	**0.677**	**0.244**	reference	reference	–
CatBoost (default)	0.676	0.245	+0.001 [−0.002, +0.004]	−0.001 [−0.005, +0.004]	No
CatBoost + class weight	0.674	0.231	+0.003 [−0.001, +0.007]	+0.014 [+0.007, +0.021]	AP only
Random Forest	0.668	0.224	+0.010 [+0.005, +0.014]	+0.020 [+0.012, +0.030]	Yes
Logistic Regression	0.667	0.218	+0.011 [+0.006, +0.016]	+0.027 [+0.017, +0.036]	Yes
Decision Tree	0.650	0.205	+0.028 [+0.020, +0.036]	+0.039 [+0.029, +0.050]	Yes

Key Results:

- Significantly better than LR, RF, DT (and their weighted variants) in AUC and AP, but small: ΔAP 0.020–0.039 (about 12% relative over LR)
- Tuning gives no detectable gain over default CatBoost
- F1 differences are mostly not significant (flat F1 optimum); recall/specificity differences mainly reflect different thresholds

Comparison with Published Results

Table 6: Published results on the same dataset family vs. this work

Study	Method	Reported metric	Split unit stated?
Sarthak et al. (2020)	DNN, embeddings	AUROC 97.4%, acc. 95.2%	Not in abstract
Goudjerkan & Jayabalan (2019)	MLP, heavy preprocessing	AUC ≈95%, recall ≈99%	Not in abstract
Emi-Johnson & Nkrumah (2025)	LR, RF, XGBoost, DNN	XGBoost best; AUC not verified	Not verified
This work	**Tuned CatBoost**	**AUC 0.677; AP 0.244**	**Yes: 0 shared patients**

Key Findings:

- Same administrative features give AUC $\approx$ 0.65–0.68 across all our algorithms, even logistic regression
- Values above 0.9 are 0.25–0.30 higher; encounter-level splits can inflate them, but we do **not** claim our model is "worse"
- External comparison is provisional until the original papers are checked

Ablation Study – Class Weighting

Table 7: Effect of class weighting on AP

Family	CV def.	CV wt.	Val def.	Val wt.
Logistic Regression	0.2091	0.2073	0.2132	0.2120
Decision Tree	0.1920	0.1899	0.2122	0.2052
Random Forest	0.2192	0.2109	0.2209	0.2071
CatBoost	0.2270	0.2161	0.2328	0.2248

Key Findings:

- Weighting did not improve AP for any family (loss: ≈ 0.002 for LR, 0.008–0.014 for RF and CatBoost)
- Weighted models are scores, not risks: best thresholds 0.53–0.62 vs ≈ 0.13 unweighted
- Threshold moving is enough to reach a good operating point (hypothesis, not tested separately)

Ablation Study – Hyperparameter Tuning

Table 8: Default vs. tuned CatBoost

Setting	**CV AP**	**Test AUC / AP**
Default (200 iter., depth 6, lr 0.1)	0.227 (3-fold)	0.676 / 0.245
Tuned (depth 5, 764 iter., lr 0.058)	0.234 (5-fold)	0.677 / 0.244

Key Findings:

- CV AP rises about 0.007 ($\approx$3%), same order as fold-to-fold std (0.005); folds differ (3 vs 5)
- On the test set: $\Delta\text{AUC} = +0.0014$ ($p = 0.355$), $\Delta\text{AP} = -0.0005$ ($p = 0.875$), both non-significant
- Performance comes mainly from the features and model family

Ablation Study – Decision Threshold

Table 9: Operating points on the test set

Operating point	Thr.	Flagged	Recall	Prec.	F1	Spec.
Tuned CatBoost, max-F1 on val (selected)	0.145	22.2%	0.431	0.217	0.288	0.804
Default CatBoost, its own max-F1 on val	0.128	28.9%	0.504	0.195	0.281	0.738
Tuned CatBoost, conventional cut-off	0.500	0.36%	0.018	0.556	0.035	0.998
Flag everybody	–	100%	1.000	0.112	0.201	0.000
Flag nobody	–	0%	0.000	–	0.000	1.000

Key Findings:

- Default 0.5 is unusable: flags 0.36% of encounters, recall 0.018 ($\approx$40 of 2,216)
- Moving to the lower threshold of default CatBoost (0.128): $+7.3$ points recall, -2.2 points precision; ΔF1 only 0.007 and not significant
- F1 optimum is flat: the threshold is a business decision (capacity), not a statistical one
- F2 caution: "flag everybody" scores F2 0.387 vs 0.360 for the model, so judge with explicit costs

Cost-Sensitive Analysis

A false alarm costs 1, a missed readmission costs r; total cost $= r \cdot FN + FP$ (hypothetical costs).

Table 10: Test set, fixed threshold 0.145

r	Nobody	Everybody	Model	Model vs best
1	2,216	17,586	4,709	+112.5%
2	4,432	17,586	5,971	+34.7%
3	6,648	17,586	7,233	+8.8%
5	11,080	17,586	9,757	−11.9%
10	22,160	17,586	16,067	−8.6%
15	33,240	17,586	22,377	+27.2%

Table 11: Validation-optimised thresholds

r	Thr.	Flagged	Saving
3	0.273	3.6%	5.5%
5	0.183	11.9%	13.0%
10	0.086	56.0%	16.4%

Theory:
$t^* = 1/(1+r) = 0.25,\ 0.167,\ 0.091.$

Key Findings:

- At the fixed threshold 0.145, the model beats both trivial policies only for $3.6 < r < 11.2$ (3447/954 and 14139/1262)
- At $r \leq 3$ the best fixed-threshold action is to intervene on nobody; at $r = 5$–10 the saving is 9–12% (test)
- Validation-optimised thresholds give 13–16% at $r = 5$–10, but these are validation estimates, not unbiased test results

Decision Curve and Review Capacity (Top-k)

Net benefit at $p_t = 0.145$

- Model: $\frac{954}{19802} - \frac{3447}{19802} \times 0.170 = 0.0187$
- Treat everyone: -0.0387; treat no one: 0
- $\approx$ 19 net true positives per 1,000 encounters

Table 12: Recall@k and lift (test)

Top k	Captured	Recall	Lift
5%	362	16.3%	3.27
10%	561	25.3%	2.53
20%	889	40.1%	2.01
30%	1,153	52.0%	1.73
50%	1,564	70.6%	1.41

Key Findings:

- Useful when follow-up capacity is fixed: top 10% of encounters captures 25% of readmissions
- Ranking value only; it does not mean probabilities are accurate for individual patients
- Full decision curve over all p_t is left as future work

Error Analysis

At threshold 0.145

- 1,262 false negatives, 3,447 false positives
- Discharged home (code 1): 12,030 encounters, prevalence 9.0%, recall only 0.305
- $\approx$ 750 of 1,262 misses ($\approx$60%) come from this single group
- Small disposition groups (codes 5 and 22, $<$ 500 encounters) show very high recall but are too small to be used as evidence

Table 13: Discharge disposition (test)

Group	Prev.	Recall	AUC
Home (1)	9.0%	0.305	0.661
Home health (6)	12.3%	0.442	0.642
Skilled nursing (3)	14.4%	0.503	0.581

Insights: Recall rises with group prevalence, so the model leans on discharge disposition as a coarse risk marker and separates risk poorly inside the large home-discharge group. No feature-importance or feature-level FN/FP analysis was run, so no further claims are made.

Calibration and Subgroup Analysis

Calibration (test)

- ECE = 0.006, slope 1.03, intercept 0.048
- Top-risk bin: predicted 0.280 vs observed 0.283
- Brier 0.0936 vs 0.0994 (constant), log loss 0.327 vs 0.351
- Better than constant, but only slightly

[Calibration curve]
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Subgroups (AUC)

Female / Male	0.685 / 0.669
Recall F / M	0.458 / 0.399
Caucasian / African Am.	0.675 / 0.672
Age 50–60	0.711
Age 80–90	0.642
Age 90–100	0.524
Diabetes / Circulatory dx	0.723 / 0.632
Discharge home / SNF	0.661 / 0.581

Near chance for age 90+ (70 positives). Female–male recall gap is 5.9 points (encounter-level 95% CI [1.8, 10.0], ignores patient clustering). All other subgroup metrics have no CIs or tests: exploratory, not evidence of bias.

Conclusions

Summary:

- We present a **leakage-safe** pipeline for 30-day readmission prediction on diabetic patients
- Combines patient-level splits, AP-based model selection, tuned CatBoost and a validation-selected threshold (0.145)
- Tuned CatBoost: ROC-AUC **0.677**, AP **0.244** (2.18× random), recall 0.431, precision 0.217, ECE 0.006

Key Technical Contributions:

1. **Rigorous evaluation**: patient-cluster CI, Holm-corrected tests, calibration, cost and subgroup analysis
2. **Honest findings**: gain over simple models is real but small; tuning and class weighting add nothing detectable
3. **Practical value**: at the fixed threshold it pays off only if a miss costs about 3.6–11.2× a false alarm (saving 9–12% on test at $r = 5$–10; 13–16% with validation-optimised thresholds); top 10% captures 25% of readmissions

Message: weak-to-moderate risk stratification only; it helps prioritise follow-up but cannot make individual-level decisions.

Limitations

- **Weak associations**: highest correlation $= 0.167$; Cramér's $V = 0.112$
- **Limited EDA**: univariate analysis; plots show counts, not readmission rates
- **Information loss**: winsorization, broad ICD-9 grouping and missing-value handling
- **Limited validation**: one patient-level split, one seed; no external or temporal validation
- **Outdated data**: 130 US hospitals (1999–2008); may not reflect current practice
- **Model limitations**: one-hot encoding and limited tuning (20 configurations); no significant improvement over default CatBoost
- **Threshold**: chosen by F1 on a model fitted on 80% of training data, then applied to the refitted model
- **Costs and subgroups**: cost ratios are assumptions; subgroup results are exploratory

Future Work

- **Robust evaluation**: repeated or nested patient-grouped CV and external validation
- **Calibration and cost**: optionally recalibrate (Platt/isotonic) and optimise thresholds using hospital-specific costs
- **Fairness**: full decision curve, patient-cluster subgroup CIs and fairness analysis
- **Richer features**: longer utilisation history, lab results, clinical notes and medication changes
- **Model improvement**: native categorical CatBoost, alternative models (LightGBM, XGBoost) and SHAP analysis